

Quiz - Lecture 04

Close your laptops and put away all electronics. No electronics or notes are allowed during the duration of this quiz.

Question 1

What is the output of this code?

```
Python
x = 5
x = x + 1
x = x * 2
print(x)
```

- A. 6
- B. 10
- C. 12
- D. 8

Question 2

What kind of error will this code produce?

```
Python
x = 10
print(x / )
```

- A. Syntax Error
- B. Semantic Error
- C. Runtime Error
- D. Functional Error

Question 3

What does the keyword `import` do in Python?

- A. It automatically runs all functions in the imported module when the program starts
- B. It loads a set of variables and functions from long-term storage into memory
- C. It installs the module from the internet so it can be used in your code
- D. It checks your code for syntax errors before the program runs

Question 4

Which of the following is an example of a Python **reserved keyword**?

Hint: Remember, reserved keywords are special words used to specify the structure of a program. Keywords can't be used as variable names.

- A. **class**
- B. string
- C. hello
- D. x

Question 5

Which of the following is a valid variable name?

- A. 1name
- B. billion!
- C. int
- D. **popular**

Question 6

What does the following code do?

```
Python
def greet():
    print("Hello")

greet
```

- A. Prints Hello
- B. Causes a syntax error
- C. **Does nothing visible**
- D. Causes a runtime error

Question 7

What is the output of this code?

Python

```
def square(x):  
    return x * x  
  
print(square(3))
```

- A. 3
- B. 6
- C. 9
- D. None of the above

Question 8

What is the output of this code?

Python

```
def add(a, b):  
    print(a + b)  
  
result = add(2, 3)
```

- A. 5
- B. "5"
- C. 5.0
- D. Runtime Error

Question 9

Which statement best describes the difference between a function **parameter** and an **argument** in Python?

- A. A parameter is the value passed into a function, and an argument is the variable inside the function
- B. A parameter is the variable listed in the function definition, and an argument is the value passed to the function when it is called
- C. Parameters and arguments are the same thing and can be used interchangeably
- D. A parameter is returned by a function, and an argument is printed by a function

Question 10

What statement best describes `msg` in the function below?

Python

```
def shout(msg):  
    print(msg.upper())
```

- E. `msg` is a reserved keyword
- F. `msg` is a parameter
- G. `msg` is a return value
- H. `msg` is an integer

Bonus Question 1

What happens when this code is run?

```
Python
def foo():
    bar()

def bar():
    print("Hello")

foo()
```

- A. Nothing happens
- B. A runtime error occurs
- C. Hello is printed
- D. A syntax error occurs

Bonus Question 2

Write a function named `print_numbers` that takes an integer argument that prints from 0 to that number. For example, `print_numbers(3)` should print the following:

```
Python
0
1
2
3
```

Hint: Don't forget to use the `print()` function. If you use the `range()` function, note that the first argument is inclusive and the second argument is exclusive.

```
def print_numbers(num):
    for i in range(0, num, 1):
        print(i)
```